

DUET

Input

Observation

Language Instruction

Robot State

VLM

Rel-Head

Smooth local control

Errors accumulate when OOD

Relative Action Manifold:

Cannot infer global path back

Robot Action

Abs-Head

Global control and OOD recovery

Limited generalization

Absolute EEF Manifold:

Global pose target provides recovery direction

Robot Action



Results in LIBERO

Scene 1 Action Disturb

Scene 2 Camera Blackout

REL $\alpha=0.04$
dis=0.28 sd=0.00 w=0.31 j=1.0cm
cfg w=0.97

ABS $\alpha=0.60$
dis=0.55 sd=0.01 w=1.00 j=10.0cm
cfg w=0.97

ABS $\alpha=0.82$
dis=0.68 sd=0.00 w=1.00 j=0.4cm
cfg w=0.40

REL $\alpha=0.08$
dis=0.33 sd=0.00 w=0.61 j=0.8cm
cfg w=0.39

REL $\alpha=0.06$
dis=0.15 sd=0.00 w=0.39 j=0.6cm
cfg w=0.94

ABS $\alpha=1.00$
dis=0.11 sd=0.00 w=0.25 j=0.6cm
cfg w=0.94

REL $\alpha=0.04$
dis=0.11 sd=0.00 w=0.25 j=0.4cm
cfg w=0.93

OOD boundary

In-Distribution: manifolds coincide

Out-of-Distribution: manifolds diverge

Recovery

Latent / State Space